

Investigating Perceived Message Credibility and Detection Accuracy of Fake and Real Information Across Information Types and Modalities.

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ABSTRACT

(Mis-)information thrives on social media, so it has become increasingly important for users to tell real from misleading content because erroneously following misinformation can cause serious consequences. In this study, we investigated users' subjective perceived information credibility and objective detection accuracy of fake and real information across three topics, each of which was delivered via two modalities. We conducted an online within-subject experiment ($n = 293$) with a three (information topics: health, science, and life) by two (modalities: text and image) by two (veracity: real and fake) design. Overall, our participants were better at identifying fake information from real information. Results also found that information types significantly mediated information modality and veracity. For real information, the text mode helped people perceive credibility in health, life, and science topics. However, for false information, the image mode helped raised the perceived credibility of life information than the text mode.

CCS CONCEPTS

• **Human-centered computing** → *Empirical studies in collaborative and social computing.*

KEYWORDS

Information modality, information topics, information veracity, perceived message credibility, detection accuracy

ACM Reference Format:

Yu chia Tseng and Chien Wen (Tina) Yuan. 2023. Investigating Perceived Message Credibility and Detection Accuracy of Fake and Real Information Across Information Types and Modalities.. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3544549.3585719>

1 INTRODUCTION

Information technologies like social media or messenger tools facilitate the circulation of information and social interactions at different scales; however, they also spawn the spread of false contents via the hyper-connected online networks [2]. Through a survey

of 1,000 individuals from a nationally representative sample in the United States, a report¹ discovered that nearly three-quarters of Americans reported to have encountered false or deceptive medical or health information and 16% found it difficult to distinguish whether a piece of information was valid or inaccurate. Depending on the topics, the widespread circulation of false information online could result in a variety of negative outcomes in health [34], political engagement [44], or daily practices based on falsehood [33].

Health-related misinformation was found to result in consequences like following ineffective or even harmful remedies and exposing oneself to health risks in critical times like the COVID-19 [11]. Levy et al. [23] discovered that false science information misled people into believing the pseudo-advantages of radioactive monazitic sand, resulting in harm to their bodies. Additionally, people may be more likely to believe false life news than other topics, as they tend to like reading news relevant to their everyday lives [36]. As people may come across (mis-)information in numerous topics every day [45], it is essential to examine whether people can discern misinformation of different topics from real information before acting upon them.

In addition to information topics, misinformation nowadays can be disseminated in various modes, including text and visual modes [12, 40]. From a decoding-encoding perspective, the text mode requires the sender to clearly convey the message and the receiver to decode the contents and sometimes even read between the lines, which opens up opportunities of communication noise and deception [41]. While textual information processing demands much efforts on the receiver end, visual components demands less decoding effort and can add context details and arouse people's emotion so that they seem more believable and attractive [7, 41]. Therefore, visual components can elicit user engagement and emotional responses (e.g., anticipation, sadness) to the information content [45]. However, when the information is false, the visual components are suggested to reveal its spurious contents, which results in lower perceived credibility [32]. Given the contemporary information landscape on social media, people can encounter a wide range of (mis-)information topics in various modes. A research gap is left unanswered regarding people's judgment of information credibility and the interaction effects between information mode (text, image), information types (health, life, and science), and information veracity (true and false).

A recent review and synthesis paper indicates that empirical research on how people encounter and process misinformation

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CHI '23, April 23–28, 2023, Hamburg, Germany
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ACM ISBN 978-1-4503-9422-2/23/04.
<https://doi.org/10.1145/3544549.3585719>

¹<https://www.goodrx.com/healthcare-access/research/medical-misinformation-survey>

is largely missing and little is known regarding how information of various styles may influence people [43]. To respond to the research gaps and contribute to the CHI community, we intend to investigate how individuals evaluate the credibility of fake and true information under various topics in both text and image modalities. Our findings intend to enrich future technological intervention designs that facilitate users to better distinguish real from fake information in domains like health, science, and life-related topics.

2 LITERATURE REVIEW

2.1 Domain-Specific Information Credibility

Information credibility can be defined as perceived quality of a piece of information from various aspects backed up by reasonable grounds [8, 13]. With credibility, it is more likely for a piece of information to be persuasive [35]. Containing empirical and verifiable evidence is a primary factor that ensures information credibility [35]. In addition to this objective aspect, information credibility relies on how its audience subjectively consider that the evidence and details can be verified. Furthermore, information credibility can be influenced by its communication channels (e.g., via face-to-face, television, or newspapers) [26] or information characteristics such as formats (e.g., text, picture, audio) that help to convey meanings to the audience [4]. For instance, Edgar found that the combination of text and image was perceived more credible than text only [10]. Interestingly, research showed that without objective verifiable contents, people can still generally evaluate the information through subjectively judging whether the story is accurate based on individual perceptions or beliefs [4]. In our study, we examined both users' subjective perceived information credibility as well as objective detection accuracy based on the veracity of the information we used in the experiment. We explored two factors that may affect perceived information credibility based on the characteristics of messages, including information types (health, life, science) and information modality (text, image), which we elaborated below.

Depending on their analytical thinking tendency as well as familiarity with a topic, individuals may find it challenging to evaluate information of various topics [30, 48]. The situation is worsened when misinformation is involved. How to distinguish real from fake news of various topics can be a critical skill in the contemporary social media landscape [47]. A prior research examined three news topics, including health, politics, and science, that drew most public interest yet also were most susceptible to misinformation, and found that people could accurately detect misinformation in politics but not so much in health and science on social media because people were exposed to political information more often [24]. It is established that misinformation has posed as the biggest threat to the society and reduced trust [1, 48]. But people encounter a mix of misinformation and real information in their daily lives, it is important to examine how people may differ when distinguishing real from fake news of various topics to offer technological support from a human-centered perspective.

We propose to examine people's capability in evaluating (mis-)information on health, science, and life topics because the consequences of believing in misinformation of these topics can jeopardize people's life and well-being [5, 14, 33]. For example, health-related misinformation has become rampant since the pandemic and

caused serious issues in public health such as vaccine hesitancy and rejection [5]. Many tools and fact-check services have burgeoned to help combat misinformation, especially for health-related issues [21]. In addition to health information, another important domain to explore is life-related news. Government agencies around the world like the Federal Trade Commission in the U.S.² or Australian Competition and Consumer Commission³ have scam watch on their websites that announce prevalent misleading misinformation, which lures people to fall victim to cases of phishing, scams, or frauds. Various types of misinformation concerning people's everyday lives thrive on the internet in forms like clickbaits, deceptive ads, or misinformation sites [22]. As people prefer to read what is most relevant to their personal lives [36], it is concerning if they believe and follow false life news. On the other hand, since the comprehension of scientific findings is more challenging and may require much literacy [6], we surmise that people may find it more difficult to evaluate the credibility and veracity of science news than other news topics. In the present study, we try to compare and contrast real and fake news across topics including health, life, and science to see how our participants' evaluation of information credibility and veracity may differ:

RQ1 : How do people's capability vary when evaluating information of various topics, including health, science, and life-related?

2.2 Information Modality and Credibility

In addition to information topics, research has shown that information modality can influence people's information processing for credibility evaluation. For example, images were found to better persuade audiences into changing their attitudes and behaviors because images can better arouse people's emotions [18]. While information in text engages people in a systematic and logical way [41], information in image can provide more details that are close to the reality and give more contexts to evoke emotions in people [18]. Miniard et al. [27] found that the participants reduced the scores of their product evaluations when the advertisements contained inappropriate or unattractive visual elements. On the contrary, if the information was presented with proper photographs and positive images, there was a significant persuasion effect on people with additional visual images added to texts [37].

In terms of information credibility, Wasike[46] found that when fake news was presented in images, people might perceive it of lower credibility than that in texts. Similar results were also found in another study [42] that image misinformation received lower perceived credibility than text one. It is less examined if people can correctly judge the credibility and veracity of real and fake information in both text and image modalities. Hence, we propose our second research question:

RQ2 : How does modality (text and image) influence people's perceived credibility on veracity of information?

To add much complexity into the issue, information credibility and veracity conveyed via different modalities may also vary across different subject matters. For example, research found that health information in text could enhance persuasiveness because the text modality can increase informational utility for promoting health

²<https://consumer.ftc.gov/consumer-alerts/2023/02/top-scams-2022>

³<https://www.scamwatch.gov.au/types-of-scams>



Figure 1: Materials in health (left), life (middle), and science (right) in text (top) and image (bottom) modes (We translated the materials for the paper; original versions were in Mandarin.)

[20]. Science information may be more suitable when presented in image because image enhances the translation and transfer of abstract scientific findings with vivid visual elements [17]. Moreover, McCabe and Castel [25] discovered that people treated science articles with images to have higher perceived credibility than those with no images. However, Gruber and Dickerson [15] found that scientific materials with images did not differ in perceived credibility from other topics such as filming and artistic domains. Given the inconsistent findings from the previous research regarding information of various topics and modalities, we propose to examine these factors in tandem by presenting both real and fake information for user evaluation with the following research question:

RQ3: How were text- and image-based information perceived as true or fake across different information types?

3 METHODS

3.1 Participants

We recruited 293 participants from different sources, including recruitment groups on Facebook and a local broadcast bulletin board (BBS). Our average participant was 31 years old ($SD = 10.09$), 192 of whom were female. 8.8% of the participants had an annual income of more than NTD\$1,000,000 (roughly USD\$33460), 6.1% NTD\$600,000–900,000 (roughly USD\$20076–30114), 39.2% NTD\$200,000–590,000 (roughly USD\$6692–19741), and 31.7% less than NTD\$200,000. Participants were compensated with a gift voucher of NTD\$100 (roughly USD\$3) for completing the online experiment.

3.2 Study Design and Stimuli

A three (information type: health, life, or science) by two (information modality: text, image) by two (veracity: true, fake) within-subject online experiment using Qualtrics was conducted. Each condition had two pieces of unique materials. All participants went through the randomly assigned twenty-four materials and the experiment lasted no more than 30 minutes. Following our institutional review board (IRB) protocol, we informed all participants that they could leave the experiment at any time.

The experiment materials were drawn from the internet such as news websites and the Taiwan FactCheck Center⁴ (see Figure 1). For the health-related materials, we included contents about food and health behaviors (e.g., “cancer can be cured by eating papaya”); for life-related materials, we included everyday issues (e.g., “According to the Consumer Protection Foundation, traffic accident injuries related to automobiles and motorcycles are not covered by insurances for medical conditions”); and science-related materials involved scientific research findings (e.g., “Scientists found that fingerprints can be created by using 2D inkjet printers, which can successfully unlock fingerprint identification systems such as phones and banking apps”).

3.3 Procedure

Participants were instructed to evaluate a collection of information at the beginning of the experiment. After agreeing to the informed

⁴<https://tfc-taiwan.org.tw/>

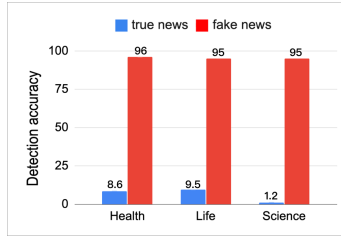


Figure 2: Detection accuracy
across different information types.

consent, the participants were randomly exposed to our materials one at a time in three different topics (health, life, or science), modalities (text or image), and veracity (true or fake). All materials were unique in content. After reading the material, the participants were asked if they had seen the information before and how much they believed the news to be true. 92% of our participants reported to have previously encountered some or all of the materials; only sixteen subjects reported that they had not seen any of the experimental stimuli. Lastly, the participants answered the demographic questions.

3.4 Measures

We used an one-item scale that measures participants' perceived credibility of the information [4]: "How much do you believe this news to be true?" Responses ranged from 1 (extremely disagree) to 5 (extremely agree) (true news: $M = 3.36$, $sd = .37$; fake news: $M = 2.49$, $sd = .55$).

We computed detection accuracy for perceived veracity using participants' credibility scores. Since perceived credibility is a continuous scale and veracity judgment (fake or true) is binary, we re-coded credibility score 1 and 2 as judgment about the information to be fake, 4 and 5 as judgment about the information to be true, 3 as undetermined. Then, if the participants labeled a piece of information as 1 or 2 on the credibility scale and the information turned out to be fake, the detection accuracy was accurate (coded as 1); if the perceived credibility of a piece of information is 4 or 5 and the information is fake, the detection accuracy failed (coded as 0). If the perceived credibility is 3, the undetermined detection accuracy was coded as 0.5. The real news was evaluated using the same coding style.

4 RESULTS

Descriptive statistics showed that average detection accuracy for fake information was 95.6% ($sd = .06$), and 7.6% ($sd = .08$) for real, suggesting participants were better at identifying fake information.

Furthermore, a one-way ANOVA showed a main effect on information types ($F(1,292) = 7.79$, $p = .006$, $\eta^2 = .02$). Post-hoc analysis revealed health information ($M = .96$, $sd = .005$) was significantly higher than life ($M = .95$, $sd = .005$, $p = .003$) and science ($M = .95$, $sd = .005$, $p = .006$) but no difference between life and science ($sd = .005$, $p = .93$). (see Figure 2). A one-way ANOVA also showed a main effect on information types ($F(1,292) = 83.95$, $p < .001$, $\eta^2 = .22$). Post-hoc analysis revealed science ($M = .01$, $sd = .001$) was

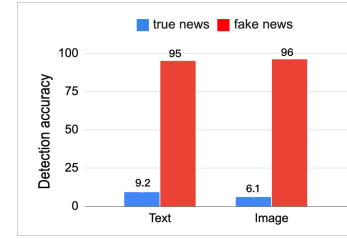


Figure 3: Detection accuracy
across different modalities.

significantly lower than health ($M = .08$, $sd = .007$, $p < .001$) and life ($M = .09$, $sd = .007$, $p < .001$), but no difference between health and life ($sd = .008$, $p = .24$).

With respect to the modality, another one-way repeated-measure ANOVA was performed to compare the effect of modality on the detection accuracy of fake news. The result showed that there was a main effect on modalities, $F(1,292) = 7.71$, $p = .006$, $\eta^2 = .02$. The post-hoc analysis revealed that image ($M = .96$, $sd = .004$, $p = .006$) was significantly higher than text ($M = .95$, $sd = .005$) (see Figure 3).

We also performed a one-way repeated-measure ANOVA to compare the effect of modality on the detection accuracy of real news. The result showed that there was a main effect on modalities, $F(1,292) = 32.13$, $p < .001$, $\eta^2 = .02$. The post-hoc analysis revealed that text ($M = .09$, $sd = .006$) was significantly higher than image ($M = .06$, $sd = .005$, $p < .001$).

To investigate our participants' perceived credibility according to different information types, a two-way repeated measure ANOVA was performed. The results suggested that there was a simple main effect in information type, $F(2,584) = 23.06$, $p < .001$, $\eta^2 = 0.7$, and veracity, $F(1,292) = 810.28$, $p < .001$, $\eta^2 = .73$. Moreover, there was an interaction between information type and veracity, $F(2,584) = 101.66$, $p < .001$, $\eta^2 = .25$ (see Figure 4).

Then, we performed a paired sample T-test to further examine the interaction effects. The result showed that for true information, the perceived credibility for science information ($M=3.11$, $sd=.03$) was lower than health information ($M=3.64$, $sd=.03$; $t(292)= -5.6$, $p < .001$) and life information ($M=3.34$, $sd=.03$; $t(292)= -12.9$, $p < .001$).

For fake information, the perceived credibility for science information was higher ($M=2.65$, $sd=.03$) than health information ($M=2.38$, $sd=.03$; $t(292)= 7.3$, $p < .001$) and life information ($M=2.43$, $sd=.03$; $t(292)= 5.7$, $p < .001$). In other words, our participants perceived the fake information to be more real but the true information to be less real, especially when the information topic is about science.

Then, to answer how different the two modalities were perceived to be credible, a two-way repeated-measure ANOVA was performed and the results suggested that there was a simple main effect in modalities, $F(1,292) = 174.73$, $p < .001$, $\eta^2 = .37$ and veracity, $F(1,292) = 810.28$, $p < .001$, $\eta^2 = .73$. Also, there was an interaction effect between modality and veracity, $F(1,292) = 203.47$, $p < .001$, $\eta^2 = .41$ (see Figure 5). A paired sample T-test was performed to further examine the interaction. The result revealed that when the

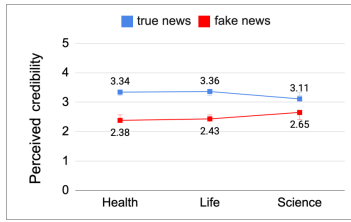


Figure 4: Interaction between information types and veracity on perceived credibility.

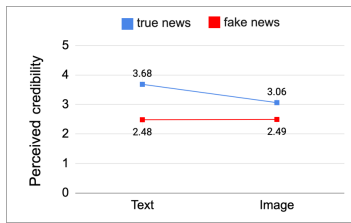


Figure 5: Interaction between modality and veracity on perceived credibility.

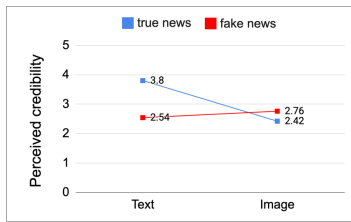


Figure 6: Interaction between modality and veracity on science information.

fake information was presented in text ($M=2.48$, $sd=.64$), it did not have significant differences than that presented in image regarding perceived credibility ($M=2.49$, $sd=.58$), $t(292)=-.39$, $p=.69$). On the other hand, when the true information was presented in text ($M=3.68$, $sd=.42$), our participants perceived it to be more credible than that in image ($M=3.05$, $sd=.51$; $t(292)=18.51$, $p<.001$).

Given that the modality had an interaction effect with veracity, we surmise that there may be a three-way interaction across different information types. A three-way repeated-measure ANOVA was performed and confirmed the significant three-way interaction effect in information type, veracity, and modality, $F(2,584)=129.55$, $p<.001$, $\eta^2=.30$. Then, we performed another two-way repeated-measure ANOVA to reveal how veracity and modality may vary across information types. The results revealed that for health information, there was no significant interaction between modality and perceived credibility, $F(1,292)=3.35$, $p=.06$, $\eta^2=.01$.

For life information, there was a significant interaction effect of modality and veracity on perceived credibility, $F(1,292)=5.53$, $p=.01$, $\eta^2=.01$. A paired sample T-test revealed that for true

information regarding life, the text mode ($M=3.78$, $sd=.72$) had higher perceived credibility than the image mode ($M=3.49$, $sd=.63$; $t(292)=5.5$, $p<.001$). On the other hand, for fake information regarding life, the text mode ($M=2.49$, $sd=.84$) also had a higher perceived credibility than the image more ($M=2.37$, $sd=.76$; $t(292)=2.2$, $p=.028$).

For science information, there was also a significant interaction effect of modality and veracity on perceived credibility, $F(1,292)=446.39$, $p<.001$, $\eta^2=.60$ (see Figure 6). For real science information, the text mode ($M=3.80$, $sd=.54$) was perceived to have higher credibility than the image mode ($M=2.42$, $sd=.80$; $t(292)=25.7$, $p<.001$). On the other hand, for fake science information, the text mode ($M=2.54$, $sd=.85$) was perceived to have lower credibility than the image mode ($M=2.76$, $sd=.69$; $t(292)=-4.5$, $p<.001$).

5 DISCUSSION

The current study examined how our participants perceived subjective (perceived credibility) and objective (detection accuracy) credibility of information across different types, modalities, and veracity. In terms of objective credibility, we found that the participants were more precise in detecting fake information than real one, which is in contrast to the previous studies where the deception bias may lead people to be aware of false information over real one [24]. We surmise that it is because misinformation is too common on social media [2] that people now develop the defense mechanisms against all incoming information by treating them as fake by default. The fact that people were less able to identify true information suggests our participants may not have the literacy nor the skills for information comprehension.

To break it down by topic and veracity, people were better at detecting true life information and health, followed by science. And the detection accuracy for fake health was than, science, and life information. We surmise that true health information can be more accurately detected is that facing recent pandemic, our participants were more aware to health-related misinformation. Also, because life news is more relatable and accessible to our participants making them difficult to detect, whereas science true information may require domain-specific literacy or knowledge to judge [36].

Last, in terms of modality and veracity, we found that people performed better for the real news on text information; on the other hand, image information could successfully reveal its veracity on fake news. This is consistent with the previous literature that posits the vivid visual elements of image mode can help users process information and increase context details for engagement [18, 41].

For true information, people perceived life as more credible than health and science. For false information, science was more credible than life and health. This may be explained by the illusory truth effect; people perceive false information as accurate after repeatedly seeing it [3, 29]. Our results indicate this is more prominent in the science domain.

Furthermore, the discrepancy between detection accuracy and perceived credibility by different topics revealed in our data implies that the mechanism of illusory truth effect does not work similarly across topics. Our findings suggest that for life and health topics, people may be good at identifying fake information based on their

experiences but they may not perceive them to be more credible. On the other hand, for information topics that require expertise, literacy, or knowledge [39], such as science, while people may be less accurate at detection, they perceive science misinformation to be more credible. Future works are encouraged to include more specific measures to operationalize relevant variables like familiarity to validate the relationships among the variables.

Interestingly, our results indicate a somewhat alarming curvilinear trend where the participants considered the fake information to be real but the true one to be less real. It is possible that people may struggle to judge scientific information due to a lack of domain-specific knowledge and experience. A lack of relevant knowledge can lead to confusion and misunderstanding [38], as science knowledge helps individuals remain informed and avoid misinformation [16]. Also, people process different science topics based on different mechanisms. For example, research showed that science literacy and education level did not reduce the perceived credibility of controversial fake science information like climate change due to contradictions to personal interests and beliefs [9, 19]. Also, when the misinformation contains jargons that people do not understand, such as "neurobabble" [28] or "pseudo-profound bullshit" [31], people may easily fall for the misinformation due to a lack of analytical thinking ability and readiness of accepting weak claims [31]. Future works should consider differentiating different types of science information to further unpack people's information processing and evaluation mechanisms.

Besides, our results showed that when the information was true, people perceived the text mode as more credible than the image mode; when the information was fake, there was no significant differences between the two modes (RQ2). While previous research suggests that image can better influence people on information processing based on perceived authenticity and effortless information processing the visual elements enable [41], our findings suggest that it may depend on information types (RQ3). For example, we found that providing text can effectively aiding people to differentiate between real news across topics in health, life, and science. In terms of fake news, there is no evident difference between text and image in health topics; the image mode can be beneficial for distinguishing fake life news; and the text mode is more useful for distinguishing fake science news.

ACKNOWLEDGMENTS

The work was supported by the National Science and Technology Council of Taiwan (NSTC Grant No. 109-2221-E-003 -010 -MY3; 110-2634-F-002 -051 -).

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